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Comparability of low-volume PM_{10} sampler with β-attenuation monitor in background air

Kaisa Salminen*, Vuokko Karlsson

Finnish Meteorological Institute, Sahaajankatu 20 E, FIN-00880 Helsinki, Finland

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Abstract

A field study was carried out to evaluate the performance of a continuous particulate monitor for measuring particulate matter PM_{10} at a low-concentration background air. The comparison was carried out using a LVS-PM_{10} reference sampler described in EN 12341:1998 Annex B1 and a β-attenuation particulate monitor Eberline FH 62 I-R. The field test site is situated in central Finland (62°35′N, 24°12′E). The experimental period was between 22 September and 5 December 2001 representing typical Finnish autumn and early winter weather conditions for the area. The comparison consisted of 42 parallel 24-h measurements. The PM_{10} mass concentration varied in the range 1.5–18 μg m^{-3} with an average of 5.1 μg m^{-3}. The particulate monitor was found to be capable of measuring low concentrations down to 2 μg m^{-3} with an estimated measurement uncertainty of ±1 μg m^{-3}. The PM_{10} concentrations measured by the monitor were close to the results of the gravimetric reference sampler giving slightly higher concentrations. The mean ratio (monitor/reference sampler) of all results was 1.06±0.12. When relative humidity was over 85% and daily temperature was below 0°C, the monitor results were systematically about 0.5 μg m^{-3} higher than the results of the reference sampler. In three cases when temperature suddenly decreased under freezing point, the concentration difference was as high as 2 μg m^{-3}. Absorption of water on the particles was the most probable reason for the concentration difference. As a part of validation the main water-soluble inorganic ions of the PM_{10} samples were analyzed by ion chromatography. They accounted for about 40% of the measured PM_{10} mass concentration and the relative proportion of sulfate was the largest. The variation of the chemical composition seemed not to have noticeable effect on the comparability of the two methods.
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1. Introduction

Epidemiological studies have indicated a strong correlation between adverse health effects and atmospheric particle concentration levels (Pope, 2000; Stieb et al., 2002). The European Union has given limit values for PM_{10} concentrations and published a standard for the determination of the PM_{10} fraction of suspended particulate matter. The 24-h limit value for PM_{10} particulate matter in Stage 1 (50 μg m^{-3}) is not to be exceeded more than 35 times a calendar year (Council Directive 1999/30/EC). In background areas PM_{10} concentrations are much lower than those in urban areas, rarely exceeding the limit values. Measurement data from background areas are, however, needed for the validation of mathematical long-range transport models. Data are also needed within EMEP (Co-operative Program for Monitoring and Evaluation of the Long-range Transmission of Air Pollutants in Europe) for the assessment of the long-range transported part of the aerosol particulate mass in Europe.

*Corresponding author. Fax: +358-9-19295403.
E-mail address: kaisa.salminen@fmi.fi (K. Salminen).

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For that purpose EMEP recommends to start PM_{10} measurements at least at one station in each country (EMEP, 2001).

Particulate monitors for measuring PM_{10} mass concentrations have been widely used and tested for performance evaluation in cities and near emission sources (Green et al., 2001; Chang et al., 2001; Allen et al., 1997; Salter and Parsons, 1999; Smith et al., 1997). At these locations air pollution limit values are relevant. Rural particulate matter concentrations have been measured mainly using manual/filter-based methods. Particulate monitors provide a continuous real-time measurement of PM_{10}, allowing the analysis of diurnal trends and the ability to track pollution episodes. Lower costs are another benefit of the automatic instruments, compared to the manual gravimetric methods.

In Finland PM_{10} particulate monitors have been compared to a gravimetric PM_{10} reference sampler according to the European standard in an urban area (Sillanpää et al., 2001). The purpose of this study was to evaluate the performance of a particulate monitor and to validate it at low background concentrations. In addition, the aim was to investigate the relationship between the β-attenuation monitor and the standardized gravimetric reference method in the northern climatic conditions and the prevailing atmospheric chemical composition. The European standard EN 12341:1998 (1998) was used as a basis for this comparison and validation study.

2. Experimental

2.1. Site description

A rural EMEP background station was selected for the field test site. The station was established in 1972 and a complete air quality monitoring program of EMEP is running continuously (major ions and trace elements in precipitation, inorganic sulfur and nitrogen compounds in particles, sulfur dioxide, nitric acid, nitrogen oxides and ozone in air). The station is located in central Finland (62°35′N, 24°12′E, 180 m above sea level). The surrounding area is sparsely populated countryside and the station itself is located in a forest glade. The nearest center is Ähtäri with 7300 inhabitants 6 km to the southwest from the station. The main local particulate emission sources are traffic, a local heating plant and agriculture. The closest major road, going through the center of Ähtäri, has an average traffic flow of about 2000 vehicles per day. The local heating plant is situated about 8 km to the southwest of the station and its annual particulate emissions are approximately 900 kg. During the field-test period between 22 September and 5 December 2001, local weather conditions were typical for those encountered in Finnish autumn and early winter.

2.2. Description of the measurement equipment

The reference sampler consisted of a LVS-PM_{10}-FH 95 KF (ESM Andersen) sampling inlet (2.3 $m^3\ h^{-1}$) and a flow control unit. Millipore Teflon filters (3.0 μm FS) were used for particle collection. The impaction surface of the inlet was cleaned and greased with a high-vacuum grease once a week. The sampling period was 24-h starting at 8:00 am local time. A direct recording instrument (β-attenuation particulate monitor Eberline FH 62 I-R) was tested as a candidate measurement method. The sampling system was heated a few degrees above the ambient temperature to prevent the condensation of water. The distance between the inlets of the two co-located measurement equipment was 1.5 m and the measurement height was 4 m above ground level.

The flow rate of the reference PM_{10} sampler was checked against a calibrated gas meter which has traceability to the national standard. The flow rate was regulated weekly at the test site. The variation of the flow rate was within $\pm 5\%$ from the nominal flow rate of 2.3 $m^3\ h^{-1}$. On average the flow rate was 1.7% below the nominal value. The flow rate of the monitor was calibrated at the beginning and at the end of the test period by using a calibrated gas meter. The variation of the nominal flow rate of 1 $m^3\ h^{-1}$ was $\pm 6\%$.

The 10-min average monitor data were compiled into 24-h data, which corresponds to the time resolution of the reference sampler. Both reference sampler and particulate monitor PM_{10} concentrations are expressed in STP conditions (temperature of 273 K and pressure of 101.3 kPa).

2.3. PM_{10} gravimetric analyses

The sample filters were stored in disposable petrislides that were placed in closed plastic bags. After the sample collection, the filters were handled and transported carefully in order to avoid contamination. Both before and after collection, the samples were weighed using a microbalance. The filters were conditioned in a laminar flow bench before the weighing. The air pressure, temperature and relative humidity of the weighing room were measured. The temperature of the weighing room during this test period was 22 ± 1°C, and the relative humidity was $50 \pm 10\%$. For the weighed masses an air buoyancy correction depending on the temperature, the relative humidity, and the pressure of the weighing room was made.

A Mettler M3 microbalance having a resolution of 0.001 mg was used for the weighing. The balance was placed inside a laminar flow bench. The calibration of the microbalance was checked at the beginning of each

weighing session and recalibrated when necessary. In addition, a standard weight representing the mass of exposed filters was weighed in each session to check the stability of the balance. The filters were discharged by using a polonium-210 radiation source when weighing. All filters were weighed at least three times. The respective readings had to agree within 3 μg to be accepted. The net masses of the PM_{10} samples varied from 0.100 mg up to 0.500 mg. Two laboratory blank filters were weighed in each weighing session. The observed mass differences of blank filters remained <0.015 mg during the whole test period.

2.4. PM_{10} chemical analyses

The main water-soluble ions of the PM_{10} samples collected by the reference sampler were analyzed using ion chromatography. The samples were extracted with 10 ml of ultra-pure water (Milli-Q). Anions (sulfate, nitrate, chloride) were separated on a Waters IC-Pak A HR anion exchange column (4.6 mm × 75 mm, 6 μm) connected to a Waters Anion quard Pak column with 0.22 μm filter. Sodium gluconate-borate eluent was used as a mobile phase. The detection was done using a conductometric detector (Waters). Cations (ammonium, sodium, potassium, calcium, magnesium) were separated on a IC-Pak C M/D cation exchange column (3.9 mm × 150 mm, 5 μm) connected to a Waters Cation quard Pak column with a 0.22 μm filter. EDTA-nitric acid eluent was used as a mobile phase. The detection was done using a conductometric detector (Waters). At the beginning of every analysis batch, the measurement range was calibrated. The calibration solutions were prepared from commercial standard solutions (Merck, Fluka). In every analysis batch 20% of the samples were synthetic internal control samples. These independent calibration check results were examined to be within the established control limits.

3. Results and discussion

3.1. Comparison of the instruments

The field comparison of the particulate monitor and the reference sampler consisted of a total of 42 parallel measurements. The results are shown in Fig. 1a. According to the reference sampler, the mass concentration varied in the range 1.5–18 μg m^{-3}, the respective average and median concentrations being equal to 5.1 and 3.6 μg m^{-3}. The average PM_{10} concentration was similar to what has been measured in rural locations at northern latitudes (Pakkanen et al., 2001; Cheng et al., 2000). The highest measured concentration at the beginning of the test period was due to long-range transported particulate matter. This episode lasted several days all around Finland and was also detected in this study at Ähtäri.

The agreement between the mass concentrations measured by the reference sampler and those measured by the particulate monitor is good. Fig. 2 shows the ratio of concentrations measured by the β-attenuation monitor and the reference method. The mean ratio of all results is 1.06 ± 0.12, which means that the monitor gives slightly higher results than the manual gravimetric method. The concentration difference varied from 0 to 2.6 μg m^{-3} being on average equal to 0.2 μg m^{-3}. The particulate monitor was in addition found to be capable of measuring low mass concentrations down to 2 μg m^{-3} with an estimated measurement uncertainty of ± 1 μg m^{-3}.

3.2. Connection to meteorological conditions

Fig. 1 shows the PM_{10} mass concentrations during the field test measured with the reference method and the particulate monitor, and the prevailing climatic conditions that were representative for the seasons concerned. The daily average temperature on the sampling days varied between -5° and $+10^{\circ}$C. The daily maximum temperature was 18°C and the daily minimum -9.5°C. The number of days without precipitation was 11 out of 42. The daily precipitation amount maximum was 4.5 mm, the average being 1.3 mm. The relative humidity was on average 85–90%. The daily wind speed calculated from 15-min averages varied between 1 and 3 m s^{-1}.

When the relative humidity was over 85% and the daily average temperature was below freezing point, the monitor results were systematically about 0.5 μg m^{-3} higher than the results of the reference sampler as seen in Figs. 1a and b. In addition, there were 3 days (22 November, 1 and 4 December) when the monitor results were about 2 μg m^{-3} higher. These differences were related to sudden temperature decrease below 0°C. When the temperature was very low and the relative humidity was high, the deliquescent point of aerosols most probably was exceeded. The inorganic aerosol particles may absorb water when the deliquescent point is exceeded (Chang et al., 2001). This leads to an overestimation of PM_{10} concentrations of the monitor compared to the manual sampler. Excluding the points in Fig. 2 when the meteorological parameters (temperature decrease, high relative humidity) were favorable for the deliquescence phenomena, the results were nearly equal and the ratio monitor/reference sampler was 1.01 ± 0.08.

3.3. Connection to chemical composition

The inorganic ion concentrations of the PM_{10} particles, collected with the reference sampler, were analyzed as a

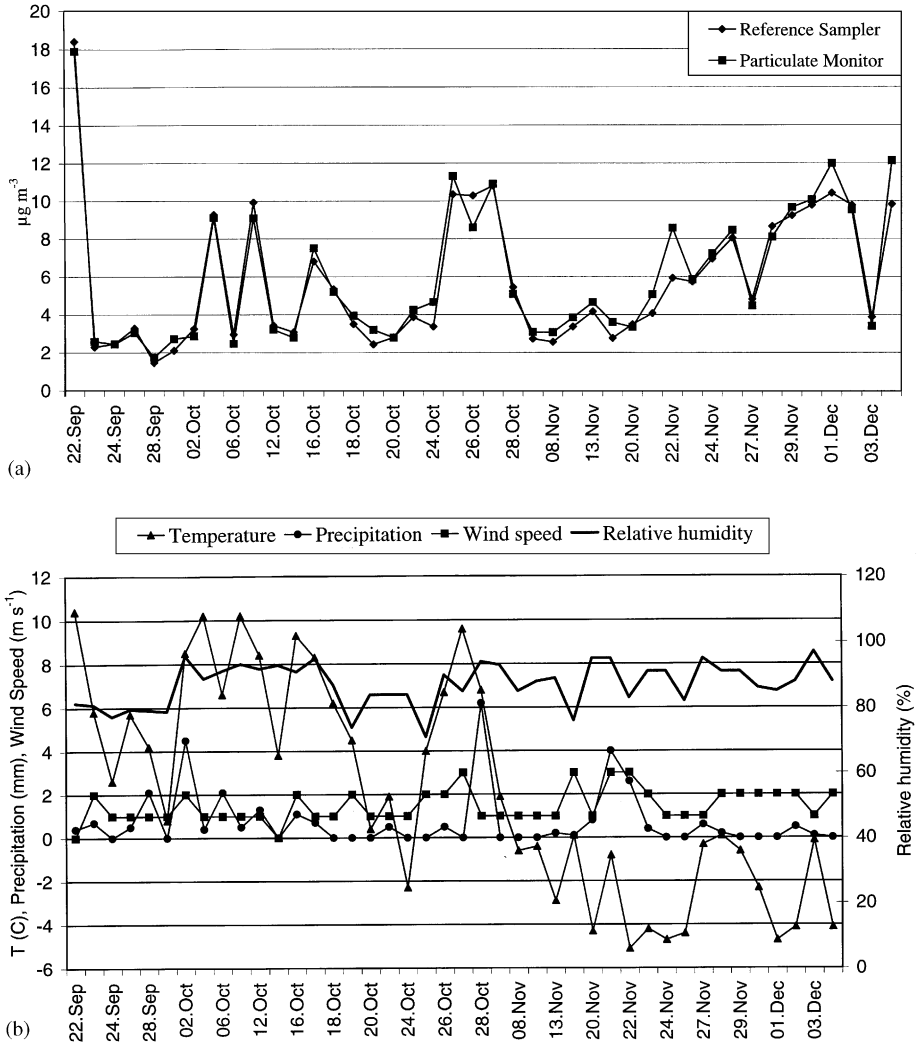


Fig. 1. (a) PM_{10} mass concentrations during the field test at Ähtäri autumn 2001. (b) Prevailing weather parameters during the field test.

part of validation conditions. The analyzed water-soluble cations (NH_4^+, Na^+, K^+, Mg^{2+}, Ca^{2+}) and anions (Cl^-, NO_3^-, SO_4^{2-}) accounted about 40% of the measured PM_{10} mass concentrations, as shown in Fig. 3. The relative proportions of sulfate, nitrate and ammonium were the largest. The fraction of sulfate varied between 8% and 49% with an average of 25%. When the daily average temperature decreased under freezing point, the relative proportion of sulfate increased. The fraction of nitrate was in the range 0.5–12% with an average of 4.5%, while for ammonium the fraction was in the range 1.5–10% with an average of 5.1%. The rest of the analyzed ions accounted for 1–35% (average 7.2%) of the measured PM_{10} mass concentrations. Similar proportions of concentrations in PM_{10} fraction have been measured in rural sites in Switzerland (Röösli et al., 2001) and in the UK (Harrison and Jones, 1995).

The Pearson correlation coefficient r between PM_{10} mass concentration and the sum of the main inorganic ions was 0.95 when the first point (22 September) was excluded from the data because of its large deviation. This sample was collected during the particulate episode and only about 20% of the PM_{10} mass concentration consisted of the main inorganic ions. The variation of the chemical composition seemed not to have an effect on the comparability of the two methods.

Particulate ammonium nitrate is a well-known labile compound causing volatilization losses and resulting in an underestimation of the PM_{10} mass. The proportion of ammonium and nitrate together accounted for about 10% of the PM_{10} mass in this field experiment. The

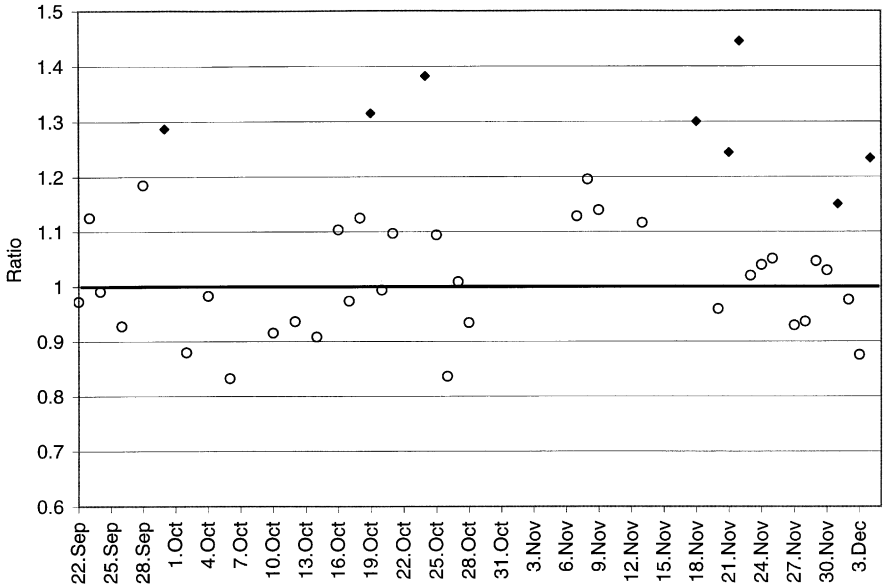


Fig. 2. Ratio of the PM_{10} concentrations measured by the β-attenuation monitor Eberline FH 62 I-R and the LVS PM_{10} gravimetric reference sampler. The mean ratio (monitor/reference sampler) of all results is 1.06 ± 0.12 and 1.01 ± 0.08 when marked (♦) results are excluded.

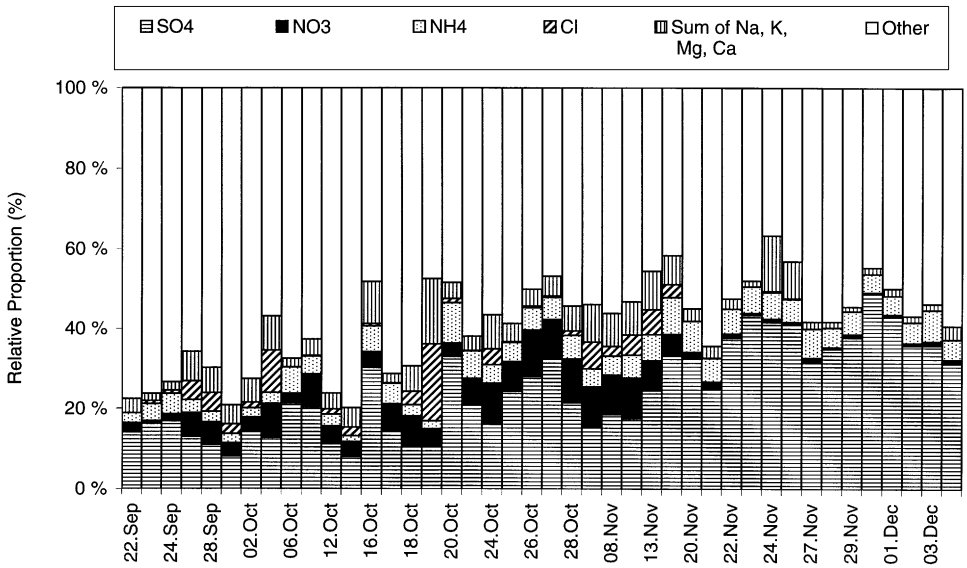


Fig. 3. Relative proportions of the main inorganic ions in PM_{10} samples.

possible evaporation losses of ammonium nitrate would be covered by the measurement uncertainty.

4. Conclusions

In this work the performance of a continuous particulate monitor for measuring PM_{10} mass at a low-concentration air was studied. The study was carried out using a β-attenuation particulate monitor and a gravimetric reference sampler. The comparison consisted of 42 parallel 24-h measurements. During the field test weather conditions represented typical Finnish autumn and early winter, with large variations in the temperature around freezing point, a high number of rainy days and a high relative humidity. The average PM_{10} mass concentration was equal to 5.1 $\mu g\ m^{-3}$ and the concentration varied in the range 1.5–18 $\mu g\ m^{-3}$.

The agreement between the PM_{10} concentrations measured by a β-attenuation particulate monitor and a gravimetric PM_{10} reference method was very good. The PM_{10} concentrations measured by the monitor were slightly higher and the mean ratio (monitor/reference sampler) of all results was 1.06 ± 0.12. When the relative humidity was over 85% and the daily temperature was below 0°C, the monitor results were systematically about 0.5 $\mu g\ m^{-3}$ higher than the results of the reference sampler. There were three cases out of 42, where the concentration difference was as high as 2 $\mu g\ m^{-3}$ and these cases were related to sudden temperature decrease below freezing point. The monitor seemed to overestimate PM_{10} concentrations, when the meteorological parameters, temperature and relative humidity, were favorable for the deliquescence phenomena and thereby for water absorption on the particles. When excluding the points, when the temperature was very low and the relative humidity was high (8 days out of 42 days) the PM_{10} concentrations were nearly equal and the ratio monitor/reference sampler was 1.01 ± 0.08.

The inorganic ion concentrations of the PM_{10} particles, collected with the reference sampler, were analyzed as a part of validation conditions. The sum of the main water-soluble inorganic ions accounted for about 40% of the measured mass concentrations, sulfate having the largest relative proportion of the analyzed ions. The average chemical composition of the PM_{10} particulate matter was similar to that measured in other rural areas in Europe. The variation of the chemical composition seemed not to have a noticeable effect on the comparability of the two methods.

This study shows that the particulate monitor is suitable for determining the PM_{10} fraction at a low-concentration background air. In addition, the particulate monitor is found to be capable of measuring low mass concentrations down to 2 $\mu g\ m^{-3}$ with an estimated measurement uncertainty of ± 1 $\mu g\ m^{-3}$.

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